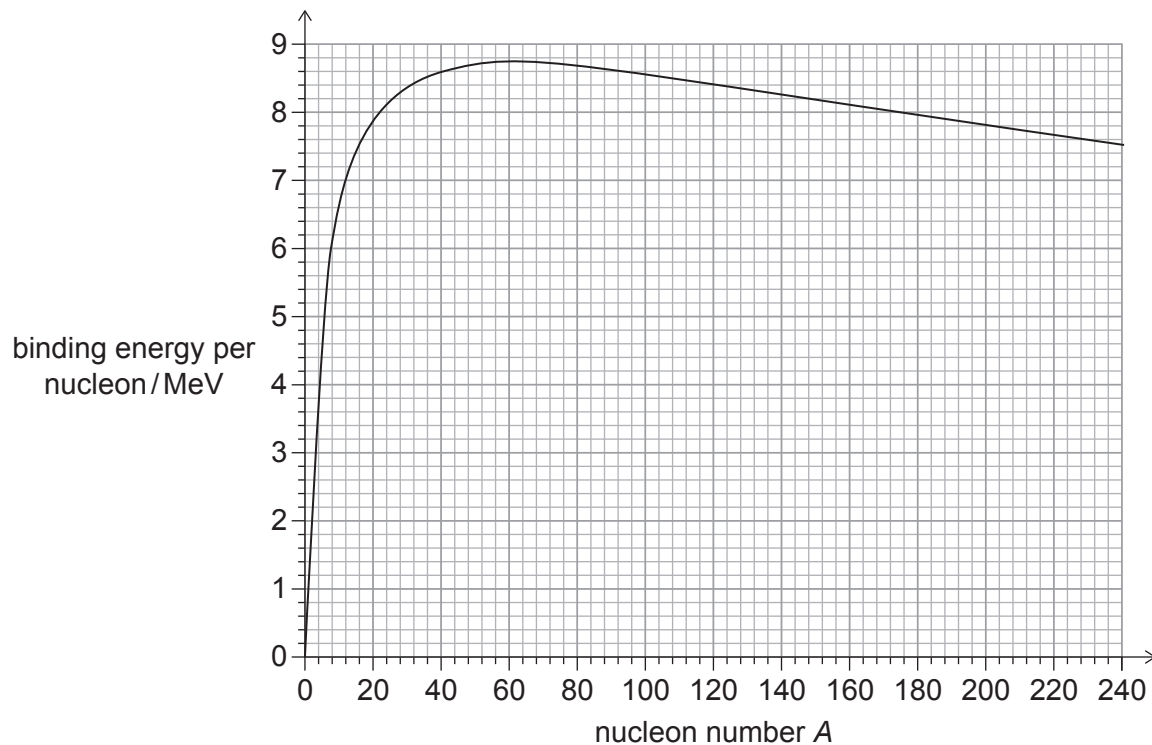


E.3 / PAPER 1 / QUESTION 7

5 November 2025 / TZ3 (file label) / HL Paper 1A / original Q37 / 8825-6719

7. A graph of the variation with nucleon number A of the binding energy per nucleon is shown.



What is the approximate total energy, in MeV, needed to completely separate the nucleons of a $^{180}_{72}\text{Hf}$ nucleus?

- A. 580
- B. 620
- C. 1440
- D. 2020